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International Institutions for Advanced AI

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Abstract

International institutions may have an important role to play in ensuring advanced AI systems benefit humanity. International collaborations can unlock AI's ability to further sustainable development, and coordination of regulatory efforts can reduce obstacles to innovation and the spread of benefits. Conversely, the potential dangerous capabilities of powerful and general-purpose AI systems create global externalities in their development and deployment, and international efforts to further responsible AI practices could help manage the risks they pose. This paper identifies a set of governance functions that could be performed at an international level to address these challenges, ranging from supporting access to frontier AI systems to setting international safety standards. It groups these functions into four institutional models that exhibit internal synergies and have precedents in existing organizations: 1) a Commission on Frontier AI that facilitates expert consensus on opportunities and risks from advanced AI, 2) an Advanced AI Governance Organization that sets international standards to manage global threats from advanced models, supports their implementation, and possibly monitors compliance with a future governance regime, 3) a Frontier AI Collaborative that promotes access to cutting-edge AI, and 4) an AI Safety Project that brings together leading researchers and engineers to further AI safety research. We explore the utility of these models and identify open questions about their viability.

Executive Summary

Recent advances in AI have highlighted the potentially transformative impacts of advanced systems.[1] International institutions may have an important role to play in ensuring these are globally beneficial.

International collaborations could be important for unlocking AI's abilities to further sustainable development and benefit humanity. Many societies that could most benefit may not have the resources, infrastructure or training to take advantage of current cutting-edge AI systems. Frontier AI development may not focus on global needs, and the economic benefits of commercial AI technologies could primarily benefit developed countries. A failure to coordinate or harmonize regulation may also slow innovation.

[1]By "advanced AI systems" we mean systems that are highly capable and general purpose.

Moreover, international efforts may also be necessary to manage the direct risks posed by powerful AI capabilities. Without adequate safeguards, some of these capabilities—automated software development, chemistry and synthetic biology research, text and video generation—may be misused by malicious actors around the world with transnational consequences. Furthermore, the significant geopolitical benefits of rapid AI development decreases the likelihood of adequate AI governance without international cooperation.

This paper describes a set of international governance functions, ranging from the distribution of frontier AI systems to the setting of safety standards, that may be needed to harness the opportunities and mitigate the risks of advanced AI. Early efforts to perform some of these functions have been undertaken by inter-governmental organizations like the Organisation for Economic Co-operation and Development (OECD), the Global Partnership on AI (GPAI), the Group of 7 (G7) Hiroshima Process, the International Telecommunication Union (ITU), as well as by private sector initiatives like the Partnership on AI, the ML Commons, and International Standards Organization (ISO) and International Electrotechnical Commission (IEC) standard-setting initiatives. But the rapid rate of AI progress suggests further institutional efforts in AI global governance could be needed.

The functions we identify could be divided in multiple ways across organizations and could involve stakeholders from the public sector, the private sector, and civil society. We group these functions into four institutional models that exhibit synergies and have precedents in existing organizations, and discuss their strengths and limitations:

- An intergovernmental **Commission on Frontier AI**[2] could establish a scientific position on opportunities and risks from advanced AI and how they may be managed. In doing so, it would increase public awareness and understanding of AI prospects and issues, contribute to a scientifically informed account of AI use and risk mitigation, and be a source of expertise for policymakers.
- An intergovernmental or multi-stakeholder **Advanced AI Governance Organization**[3] could help internationalize and align efforts to address global risks from advanced AI systems by setting governance norms and standards, and assisting in their implementation. It may also perform compliance monitoring functions (either independently or in association with industry groups) for an international governance regime.
- A **Frontier AI Collaborative**[4] could promote access to advanced AI as an international public-private partnership. In doing so, it could help underserved societies benefit from cutting-edge AI technology and promote international access to AI technology for safety and governance objectives.
- An **AI Safety Project**[5] could bring together leading researchers and engineers, and provide them with access to computing resources and advanced AI models for work on technical mitigations of AI risks, potentially working with parallel industry efforts. It would promote AI safety R&D by increasing its scale, resourcing and coordination.

There are important open questions around the viability of such models. A Commission on Frontier AI will face significant scientific challenges given the limited scientific research on advanced AI

[2]Similar institutions include the Intergovernmental Panel on Climate Change (IPCC), the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES), and the Scientific Assessment Panel of the United Nations Environment Programme.

[3]Cf. the International Civil Aviation Organization (ICAO), the International Atomic Energy Agency (IAEA) and the Financial Action Task Force (FATF).

[4]Cf. international public-private partnerships like Gavi, the Vaccine Alliance and the Global Fund to Fight AIDS, Tuberculosis and Malaria; as well as organizations that hold dangerous technologies, like the IAEA's nuclear fuel bank and the Atomic Development Authority proposed following WWII.

[5]Cf. scientific collaborations like the European Organization for Nuclear Research (CERN) and ITER.

issues, and will have to combat the politicization of its activities. The rapid rate of AI progress will make it difficult for an Advanced AI Governance Organization to set standards that keep up with and are appropriately scoped for the risk landscape, and the many difficulties of international coordination raise questions about how participation in an Organization regime can be incentivized. The potentially dual-use nature of general purpose AI technologies might hamper a Frontier AI Collaborative's ability to distribute beneficial systems widely, and the significant obstacles to underserved societies making use of AI systems raises questions about its effectiveness as a means of promoting sustainable development. Finally, an AI Safety Project could struggle to secure adequate model access to conduct safety research, and it may not be worthwhile to divert safety researchers away from frontier labs.

Table 1 below summarizes our mapping of institutional functions, the challenges they address, and the existing and possible institutions that perform those functions.

There are details about these institutional models that we leave out of scope or that remain uncertain—whether these institutions should be new or evolutions of existing organizations, whether the conditions under which these institutions are likely to be most impactful will obtain, whether other groupings of institutional functions would be more effective—but rapid progress on these topics will help prepare for the development of advanced AI.

Table 1: A mapping of international institutional functions, governance objectives and models.

Function →	**Science and Technology Research, Development and Diffusion**				**International Rulemaking and Enforcement**			
Objective / institutions ↓	Conduct or Support AI Safety Research	Build Consensus on Opportunities and Risks	Develop Frontier AI	Distribute and Enable Access to AI	Set Safety Norms and Standards	Support Implementation of Standards	Monitor Compliance	Control Inputs
Spreading Beneficial Technology	No	Yes	Maybe	Yes	No	No	No	No
Harmonizing Regulation	No	No	No	No	Yes	Yes	No	No
Ensuring Safe Development and Use	Maybe	Yes	Maybe	Maybe	Yes	Yes	Maybe	Maybe
Managing Geopolitical Risk Factors	No	No	Maybe	Maybe	No	No	Yes	Yes
Existing Int'l Institutional Efforts		OECD, GPAI, G7, ITU			ISO/IEC			Semi-conductor Export Controls
Possible Institution	*AI Safety Project*	*Commission on Frontier AI*	*Frontier AI Collaborative*		*Advanced AI Governance Agency*			
Key challenges	Model access; diverting talent	Politicization; scientific challenges	Managing dual-use technology; education, infrastructure and ecosystem obstacles		Incentivizing participation; quickly changing risk landscape; maintaining appropriate scope			

1 Introduction

The capabilities of AI systems have grown quickly over the last decade. Employing a growing wealth of algorithmic insights, data sources and computation power, AI researchers have created systems that can comprehend language, recognize and generate images and video, write computer programs and engage in scientific reasoning. If current trends in AI capabilities continue, AI systems could have transformative impacts on society.

Powerful AI systems will bring significant benefits and risks. These systems have the capacity to significantly improve the productivity of workers and economies as a whole, and to help us address some of our most important social and technological challenges. But these systems also present challenges including workforce dislocation, lack of transparency, biased outcomes, inequitably shared benefits and threats to national security.

Promoting AI benefits and managing AI risks both have domestic and international components. On the domestic front, governments and the private sector will need to establish rules and norms around how advanced AI systems are developed, distributed, deployed and accessed, addressing issues like security, distributive impacts, privacy, bias, and more. A number of challenges have the potential to transcend national borders and impact societies and economies worldwide. Accordingly, policymakers, technologists, and AI governance experts have recently begun to call for specific global AI governance initiatives centered on international institutions.[6]

This paper contributes to these early conversations by discussing why AI governance may be needed on an international scale and then offering a non-exhaustive taxonomy of the institutional functions that international efforts might require. It explores four possible international institutions to perform these functions: 1) a Commission on Frontier AI that facilitates expert consensus on opportunities and risks from advanced AI; 2) an Advanced AI Governance Organization that sets international standards, supports their implementation, and could monitor compliance to future governance regimes; 3) a Frontier AI Collaborative that develops and distributes cutting-edge AI; and 4) an AI Safety Project that brings together exceptional researchers, engineers and compute to further AI safety research. Each of these approaches seeks to mitigate the societal challenges of advanced AI in different ways and each confronts significant challenges to its viability and success.

2 The Need for International Governance

Powerful AI systems have the potential to transform society, economics and politics in fundamental ways. Because of characteristics like its high barriers to development/utilization and the possibility of cross-border use, it is possible that harnessing AI's potential for global benefit and managing its risks could require governance functions at the international level.

Promoting Global Benefits

Access to appropriate AI technology might greatly promote prosperity and stability [50], but the benefits might not be evenly distributed or focused on the greatest needs of underrepresented communities or the developing world. Inadequate access to internet services, computing power, or availability of machine learning training/expertise will also hinder certain groups' ability to benefit fully from AI advances.

International institutions have long sought to support sustainable global development. International efforts to ***build consensus on AI opportunities***—especially addressing barriers to their effective use globally—could support efforts to ***distribute and enable access to AI.*** On top of facilitating access, this could also include building capacity to benefit from AI through education, infrastructure, and local commercial ecosystems. In some cases, international collaborations (including public-private

[6] See, e.g., [10; 11; 13; 22; 25; 33; 46]

partnerships) to ***develop frontier AI systems*** that are suited to the needs to underserved communities may also be appropriate.

Inconsistent national regulations could also slow the development and deployment of AI, as developers of powerful AI technology may be unwilling to export to countries with inconsistent or unsuitable technology governance.[7] International efforts to ***set safety norms and standards*** could help coordinate governance in a way that supports innovation and serves a broad set of interests.

Managing Shared Risks

Advanced AI capabilities may also create negative global externalities. AI systems today are already capable of not just progressing drug discovery and development, but also of (re)inventing dangerous chemicals [48] and solving foundational problems in synthetic biology [24]. Scientific capabilities like these could be weaponized by malicious actors for use worldwide. AI may also be used to create potent cyberweapons that can generate code, scan codebases for vulnerabilities, and engineer polymorphic malware in ways that threaten critical infrastructure [38; 41]. Existing AI systems already pose mis- and dis-information issues, and the introduction of more advanced systems is leading malicious actors to explore more sophisticated methods of information warfare.[8] Furthermore, building systems that act as intended in novel circumstances is a challenging problem that may only grow more difficult [3; 20]. As systems get increasingly capable, there will be greater incentives to deploy them in higher stakes domains where accidents could have serious global consequences [5; 9].

Implementing protocols for responsible development and deployment will help address these risks of accident and misuse,[9] on top of measures targeted at specific downstream issues.[10] However, cross-border access to AI products and the cross-border effects of misuse and accidents suggests that national regulation may be ineffective for managing the risks of AI even within states. States will inevitably be impacted by the development of such capabilities in other jurisdictions.

To further the international adoption of safety protocols for advanced models, it would be useful to ***build consensus on risks*** and how they can be mitigated, and ***set safety norms and standards*** and ***support their implementation*** to help developers and regulators with responsible development and use. International efforts to ***conduct or support AI safety research*** may be beneficial, if it can increase the rate of safety progress or the reach of its outputs.

In the longer term, continued algorithmic and hardware progress could make systems capable of causing significant harm accessible to a large number of actors, greatly increasing the governance challenges.[11] In this case, the international community might explore measures like ***controlling AI inputs*** (although the dual-use/general purpose nature of the technology creates significant tradeoffs to doing so) and ***developing and/or enabling safe forms of access to AI***.

The significant geopolitical benefits of AI development may disincline states to adequately regulate AI: arguments about national competitiveness are already raised against AI regulation,[12] and such pressures may strengthen alongside AI progress. We may eventually need international agreements

[7] In addition to compliance costs, they may be concerned about enabling misuse, or the theft of proprietary information [6].

[8] See, e.g, [7; 17]

[9] Potential protocols include: training methods that restrict the dangerous capabilities and increase the reliability of systems, subjecting AI systems to risk assessments that ascertain their propensity to cause harm before training or deployment, deployment protocols that secure systems against misuse or the exfiltration of its parameters, post-deployment monitoring to identify and respond to unforeseen risks. See [4].

[10] Such as the Digital Services Act for disinformation, and treaties targeting chemical and biological weapons.

[11] According to [14] and [21], the compute costs of training a model of a fixed performance level decreases approximately tenfold every 2 years.

[12] See, e.g., the arguments discussed in [44]

that address these geopolitical risks, with institutions that can ***monitor compliance*** where feasible.[13] Efforts to control AI inputs may be useful to enable non-proliferation of potentially dangerous capabilities and increase the technical feasibility of monitoring. More speculatively, efforts to develop frontier AI collectively or distribute and enable access and its benefits could incentivize participation in a governance regime.

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The institutional functions identified above can be summarized and grouped into the following two broad categories.

I. Science and Technology Research, Development and Diffusion

- **Conduct or support AI safety research:** Research and develop of measures to reduce the risks of AI misuse and accidents stemming from system characteristics like dangerous capabilities and unreliability/misalignment. This includes work on understanding and evaluating these characteristics and the threats they pose, training methods to reduce and manage risky behaviors, and examining safe deployment protocols appropriate to different system [3; 20].
- **Build consensus on opportunities and risks:** Further international understanding of the opportunities and challenges created by advanced AI and possible strategies for mitigating the risks.
- **Develop frontier AI:** Build cutting-edge AI systems.
- **Distribute and enable access to cutting edge AI:** Facilitate access to cutting-edge systems and increase absorptive capacity through education, infrastructure, and support of the local commercial ecosystem.

II. International Rulemaking and Enforcement

- **Set safety norms and standards:** Establish guidelines and standards around how AI can be developed, deployed and regulated to maximize benefit and minimize risks.
- **Support implementation of standards:** Provide assistance for the implementation of established guidelines and standards.
- **Monitor compliance:** Conduct audits /evaluations and issue certifications / licenses to ensure adherence to international standards and agreements.
- **Control AI inputs:** Manage or monitor models, compute, data and other ingredients of potentially dangerous technologies.

International bodies already perform some of these functions.[14] The OECD's AI Principles and AI Policy Observatory work, the ITU's AI for Good initiative, and expert reports from the Global Partnership on AI's Working Group on Responsible AI are early efforts at building consensus on AI opportunities and risks. Relatedly, the UK's proposed Foundation Model Taskforce [18] and the US's proposed Multilateral AI Research Institute could emerge as multilateral efforts to conduct AI safety research, or potentially even develop frontier AI systems, though both are in exploratory phases.[15]

Alongside lawmaking efforts like the EU's AI Act and the Council of Europe's Convention on AI, Human Rights and Democracy, we have seen early norm and standard setting efforts from ISO/IEC, but little in the way of implementation support, oversight or certification. In terms of controlling

[13] Monitoring can vary significantly in intrusiveness and effectiveness: while it will be highly difficult to implement adequate monitoring across major geopolitical divides, the safety of advanced systems could be a shared interest of major powers and a regime to address risk factors from smaller-scale geopolitical competition may be feasible.

[14] See [49] for a more thorough overview.

[15] The amount of funding required to stay on the cutting-edge of AI capabilities is significant. See, e.g., [39].

dangerous inputs: computing resources have been targeted by US, Japanese and Dutch export controls that prevent the sale of certain AI chips and semiconductor manufacturing equipment to China [1].

3 International Institutions for Advanced AI

We have outlined several AI governance functions that might be needed at an international level, and shown that only a limited number of these are currently being performed by existing institutions. In this section, we discuss how functional gaps may be filled.

The functions could be split in multiple ways across institutions: drawing on existing international organizations and proposals, we describe four idealized models. We note that the models described in this section describe roles that could be filled by existing or new institutions. Participants in these institutions could include governments, non-governmental organizations, the private sector, and academia. Table 1 summarizes the previous analysis and the functions of the institutions we discuss.

Table 1 (repeated): A mapping of international institutional functions, governance objectives and models.

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Existing Int'l Institutional Efforts		OECD, GPAI, G7, ITU			ISO/IEC			Semi-conductor Export Controls
Possible Institution	*AI Safety Project*	*Commission on Frontier AI*	*Frontier AI Collaborative*		*Advanced AI Governance Agency*			
Key challenges	Model access; diverting talent	Politicization; scientific challenges	Managing dual-use technology; education, infrastructure and ecosystem obstacles		Incentivizing participation; quickly changing risk landscape; maintaining appropriate scope			

3.1 Commission on Frontier AI: Fostering Scientific Consensus

There have been several recent proposals of an intergovernmental body to develop expert consensus on the challenges and opportunities presented by advanced AI.[16] Existing institutions like the Intergovernmental Panel on Climate Change (IPCC), the Intergovernmental Science-Policy Platform

[16] See, e.g., [27; 33].

on Biodiversity and Ecosystem Services (IPBES) and the Scientific Assessment Panel (SAS), which studies ozone depletion under the Montreal Protocol, provide possible models for an AI-focused scientific institution. Like these organizations, the Commission on Frontier AI could facilitate scientific consensus by convening experts to conduct rigorous and comprehensive assessments of key AI topics, such as interventions to unlock AI's potential for sustainable development, the effects of AI regulation on innovation, the distribution of benefits, and possible dual-use capabilities from advanced systems and how they ought to be managed.

Functions and Motivation

International consensus on the opportunities and risks from advanced AI has the potential to facilitate effective action addressing them, for example, by engendering a shared desire for the development and adoption of effective risk mitigation strategies.

Currently, there is significant disagreement even among experts about the different opportunities and challenges created by advanced AI,[17] and this lack of consensus may worsen over time as the effects of AI systems increase in scale and number, hindering collective action on the scale necessary to ensure that AI is developed for the benefit of all. Furthermore, there are several challenges from advanced AI that may require international action *before* risks materialize, and the lack of a widely accepted account or even mapping of AI development trajectories makes it difficult to take such preparatory actions. Facilitating consensus among an internationally representative group of experts could be a promising first step to expanding our levels of confidence in predicting and responding to technological trends.

Challenges and Risks

Scientific challenges of understanding risks on the horizon: Understanding frontier AI risks and their mitigation is technically challenging. The nature of future AI capabilities and their impact is difficult to predict, especially given the fast rate of progress. To increase chances of success, a Commission should foreground scientific rigor and the selection of highly competent AI experts who work at the cutting edge of technological development and who can continually interpret the ever-changing technological and risk landscape.

Unfortunately, there is a relative lack of existing scientific research on the risks of advanced AI.[18] To address the lack of existing scientific research, a Commission might undertake activities that draw and facilitate greater scientific attention, such as organizing conferences and workshops and publishing research agendas. It may be helpful to write a foundational "Conceptual Framework"—following the example of the IPBES—to create a common language and framework that allows the integration of disparate strands of existing work and paves the way for future efforts [12].

Politicization: A Commission on Frontier AI would benefit from, if not require, a clear buffer between experts charged with developing consensus narratives around the risks and opportunities of AI and policymakers acting on the political and economic interests of their states, which might push policies in different directions. The scientific understanding of the impacts of AI should ideally be seen as a universal good and not be politicized.

Membership structure can affect a Commission's impartiality and legitimacy: ideally, there would be broad geographic representation in the main decisionmaking bodies, and a predominance of scientific experts in working groups.[19] Unfortunately, given the uncertain and controversial nature of advanced

[17] See, for example, the disagreement around whether advanced AI could pose an extinction risk: [9; 31].

[18] The recent IPCC assessment, for reference, was written by 234 scientists from 66 states and drew on 14,000 scientific papers.

[19] If legitimacy is the primary concern, the Commission might adopt the IPCC's innovation of writing key documents by consensus, balancing inclusion (states' representatives and scientists review, discuss and approve the report line by line) and scientific rigor (all suggested amendments must be consisted with working group's scientific report that is being summarized) [36].

AI risks and opportunities, representation may trade off against a Commission's ability to overcome scientific challenges and generate meaningful consensus.[20] In addition to striking the correct balance in membership, a Commission should carefully scope the subject matter of their research—it may, for example, adopt the IPCC's objective of being "policy-relevant" without being "policy-prescriptive."

Assessment

The objective of a Commission on Frontier AI is worthwhile in most circumstances, but the scientific challenges and potential of politicization imply that a Commission—especially one that aims at broad political representation—may not be able to build scientific consensus effectively. The extraordinary pace of technological change may require more nimble policy responses, such as less institutionalized and politically authoritative scientific advisory panels on advanced AI.

3.2 Advanced AI Governance Organization: Promoting Norms and Standards, Providing Implementation Support, Monitoring Compliance

As discussed above, certain misuse and accident risks of advanced AI systems may pose significant global threats, and international efforts aimed at managing these risks could be worthwhile. An intergovernmental or multi-stakeholder organization could perform a variety of governance functions furthering the regulation of such systems, in particular norm and standard setting, implementation assistance, and perhaps monitoring compliance with governance frameworks.[21]

Functions and Motivation

We identify two main objectives for an Advanced AI Governance Organization. How much emphasis it should place on each depends on the challenges it aims to address.

Objective 1: Internationalizing and harmonizing AI regulation. Regulatory regimes that set standards and provide implementation support may help ensure that powerful AI capabilities do not pose misuse or accident risks. Standard setting would facilitate widespread international adoption by: 1) reducing the burden on domestic regulators to identify necessary safety regulations and protocols, 2) generating normative pressure for safety protocol adoption, and 3) reducing frictions around the development of international frameworks. Implementation support would assist the establishment and maintenance of regulatory regimes meeting these frameworks. Examples of organizations that perform similar functions include the Financial Action Task Force (FATF), the International Telecommunication Union (ITU) and the International Civil Aviation Organization (ICAO).

The same functions are useful for harmonizing regulation: international standard setting would reduce cross-border frictions due to differing domestic regulatory regimes. (It is possible that future regulations will limit access to powerful AI technologies in jurisdictions with inadequate AI governance.) Implementation support would help reduce obstacles to countries meeting international standards and therefore enable greater access to advanced AI.

Objective 2: Monitoring compliance. Where states have incentives to undercut each other's regulatory commitments, international institutions may be needed to support and incentivize best practices. That may require monitoring standards compliance. At the least intrusive end of the spectrum is self-reporting of compliance with international standards (as in the Paris Agreement—see proposals for self-reporting/registration of training runs [4]). Organizations like the FATF, ICAO, and the International Maritime Organization (IMO) take a somewhat more intrusive approach, monitoring jurisdictions to ensure they adopt best practice regulations, and in some cases checking on the enforcement of domestic regulations embodying international standard. In the case of advanced

[20] If it follows the IPCC model, experts will be nominated by member states, but there will not be a robust climate science discipline to buffer against political interests.

[21] See, e.g., [2; 10; 13; 28; 46] for discussions and proposals of an institution of this type.

AI, some observers have asked whether more intrusive forms of international oversight might be necessary, including detection and inspections of large data centers (partly analogous to IAEA safeguards). The more intense and intrusive any monitoring, the more challenging it may be to get to consensus [35; 43].

Challenges and Risks

Speed and comprehensiveness in standard setting: One challenge for a Governance Organization is that standard setting (especially in an international and multistakeholder context) tends to be a slow process, while the rapid and unpredictable nature of frontier AI progress may require more rapid international action. A Governance Organization may need to partner with faster-moving expert bodies and expedited standard-setting approaches. The breadth of membership may also represent a trade-off between speed and diversity of perspectives. Broader membership may be important where long-term consensus is important, while urgent risks may need to be addressed at first by smaller groups of frontier AI states, or aligned states with relevant expertise.

Incentivizing participation: The impact of a Governance Organization depends on states adopting its standards and/or agreeing to monitoring. Broad agreement (or agreement among frontier AI states at least) about the risks that standards and monitoring address and financial and technical support for standards' implementation may help induce states' participation. Many states—even those that are not full members of the organization—adopt FATF standards because they view them as in their own interests [45]. Other AI-specific incentives for participation include conditioning on participation access to AI technology (possibly from a Frontier AI Collaborative) or computing resources.[22] States might also adopt import restrictions on AI from countries that are not certified by a Governance Organization—similar, for instance, to the way states prohibit flights from jurisdictions without ICAO-certification from entering their airspace.

In the more distant case of high stakes agreements governing AI development by states (such as arms control treaties), some states may be especially reluctant to join due to fear of clandestine noncompliance by other states. They may also worry that international inspections could compromise state secrets to the benefit of adversaries (which information security protocols could address in part). Again, the current reliance of advanced AI development on significant computing resources may make it easier to track significant AI efforts.[23] Automated (even AI-enabled) monitoring may allow closer inspection of large training runs without compromising secrets. Such measures would likely hinge on negotiated verification regimes rather than national technical means—and negotiating verification is always fraught (e.g., in the case of the Chemical Weapons Convention) and often unsuccessful (e.g., in the case of the Biological Weapons Convention)[15; 23].

Scoping challenges: Unlike many other technologies—from nuclear resources to aviation—AI is already broadly deployed and used by billions of people every day. To operate efficiently and at appropriate scale, a Governance Organization should focus primarily on advanced AI systems that pose significant global risks, but it will be difficult in practice to decide on the nature and sophistication of AI tools that should be broadly available and uncontrolled versus the set of systems that should be subject to national or international governance. The rapid evolution of these technologies compounds the problem, as the technological frontier is advancing quickly, and models that were "frontier" a year ago are now both outdated and widely available.

[22] The cloud compute industry and the underlying semiconductor supply chain are concentrated in a small number of countries.

[23] Oversight of data centers may allow the detection of large training runs that are subject to international controls. See [8; 35].

Assessment

If advanced AI poses misuse and accident risks of a global scope, and unilateral technical defenses are not sufficient to protect against them, an international Governance Organization may be valuable. However, its effectiveness will depend on its membership, governance and standard-setting processes.

It may be important for governance to apply to all countries, and particularly to those whose firms are on the frontier of AI development. Yet, aligned countries may seek to form governance clubs, as they have in other domains. This facilitates decision-making, but may make it harder to enlist other countries later in the process. It is unclear what institutional processes would satisfy the demands of legitimacy and effectiveness, and incentivize the participation of important groups of stakeholders.

3.3 Frontier AI Collaborative: Enabling International Access to AI

Policymakers and pundits have also proposed collaborations to develop and distribute cutting-edge AI systems, or to ensure such technologies are accessible to a broad international coalition [22; 25; 34; 40]. Given the significant cost of developing advanced AI systems, a Frontier AI Collaborative could take the form of an international private-public partnership that leverages existing technology and capacity in industry, for example by contracting access to or funding innovation in appropriate AI technology from frontier AI developers. Such an organization could draw inspiration from international public-private partnerships like Gavi - the Vaccine Alliance or The Global Fund to Fight AIDS, Tuberculosis and Malaria; as well as international organizations that hold and control powerful technologies, like the IAEA's nuclear fuel bank [32] or the Atomic Development Authority that was proposed following WWII [51].

Functions and Motivation

A Frontier AI Collaborative could be designed to spread beneficial technology or serve as a channel for legitimate international access to advanced AI.

Spreading beneficial technology: A Collaborative could be established to ensure the benefits of cutting-edge AI reach groups that are otherwise underserved by AI development. One motivation for this objective is that the resources required to develop advanced systems make their development unavailable to many societies. This may result in technologies being inadequately designed for and supplied to groups that may benefit most from them for a variety of reasons:

1. Systems developed by private actors may not adequately cater to all societies or demographics: they may not reflect the right values, have the right language capabilities, or work efficiently in diverse geographies[30].
2. Private firms may not price their products in ways that allow for equitable or broad distribution of benefits.
3. In order to protect proprietary information, private AI firms may not grant deep access to their models (e.g. they may restrict API access to prevent model imitation [42]), which could preclude the development of use cases with significant social benefit.

A Collaborative could acquire or develop and then distribute AI systems to address these gaps, pooling resources from member states and international development programs, working with frontier AI labs to provide appropriate technology, and partnering with local businesses, NGOs, and beneficiary governments to better understand technological needs and overcome barriers to use.[24] It could enable the development of technology that better caters to the underserved, [30] price access to AI models

[24]For example, Gavi promotes immunization e.g. by funding innovation, and negotiating bulk contracts with pharmaceutical companies (especially advanced market commitments) for vaccination programs in low-income countries [16].

in a way that is equitable, provide education and build infrastructure to allow the effective utilization of AI technology, and set a paradigm for responsible and inclusive AI development. By pooling the resources of multiple parties towards these ends (including safety talent, which is currently very scarce in the AI community), one or more of the aims could potentially be pursued more quickly and effectively than under the status quo.

Facilitating legitimate international access to powerful AI: More speculatively, a sufficiently ambitious, responsible and legitimately governed AI Collaborative could further AI governance objectives and reduce geopolitical instability amidst fierce AI competition among states. For example, membership in a Collaborative and access to its safe technology could be offered as an incentive for countries to participate in a governance regime that enforces responsibility (such as agreements to enact stricter regulation, or restrict military AI development). The existence of a technologically empowered neutral coalition may also mitigate the destabilizing effects of an AI race between states, by reducing the strategic consequences of one party falling behind or moderating the power concentrated among competing powers.

In addition, the Collaborative's technology could be used to increase global resilience to misused or misaligned AI systems by giving experts a head start in studying the kinds of threats likely to be posed by other AI systems, and by being deployed for "protective" purposes such as fixing security vulnerabilities in critical infrastructure, detecting and counteracting disinformation campaigns, identifying misuse or failures of deployed systems, or monitoring compliance with AI regulations. This would be especially useful in scenarios where sharply falling training costs (due to algorithmic progress and Moore's law) means the ability to train dangerous models is widely spread.

Challenges and Risks

Obstacles to benefiting from AI access: It is likely to be difficult to meaningfully empower underserved populations with AI technology, as the obstacles to their benefiting from AI run much deeper than access alone. Any Collaborative whose primary objective is global benefit needs to be adequately integrated into the global development ecosystem and set up with significant capacity or partnerships for activities beyond AI development such as: understanding the needs of member countries, building absorptive capacity through education and infrastructure, and supporting the development of a local commercial ecosystem to make use of the technology [29; 47]. The resources required to overcome these obstacles is likely to be substantial, and it is unclear whether such a Collaborative would be an effective means of promoting development.

Diffusion of dual-use technologies: Another challenge for the Collaborative would be managing the risk of diffusing dangerous technologies. On the one hand, in order to fulfill its objectives, the Collaborative would need to significantly promote access to the benefits of advanced AI (objective 1), or put control of cutting-edge AI technology in the hands of a broad coalition (objective 2). On the other hand, it may be difficult to do this without diffusing dangerous AI technologies around the world, if the most powerful AI systems are general purpose, dual-use, and proliferate easily.[25] This is especially the case if the Collaborative aims to deploy cutting-edge general purpose systems to manage AI risks: the kinds of systems (and their underlying source code and algorithms) capable of meaningfully protecting against dangerous AI or furthering governance objectives may pose an exceptional misuse risk, as they will likely be engineered from highly capable, general purpose models.

To address such a challenge, it would be important for the Collaborative to have a clear mandate and purpose. Members of a Collaborative would need to have a strong understanding of the risks of the models being developed now and in the future, and their implications for model distribution, organization security (especially restrictions on the movement of Collaborative model weights), and other activities that may impact their ability to benefit from the Collaborative. Only by doing this

[25] For example: it may be difficult to protect broadly-deployed models from imitation, and it may be difficult to secure the deployment pipeline from attempts to copy model weights.

would the Collaborative be able to consistently implement the necessary controls to manage frontier systems. It may also be necessary to exclude from participation states who are likely to want to use AI technology in non-peaceful ways, or make participation in a governance regime the precondition for membership.

Assessment

A Frontier AI Collaborative may indeed be a viable way of spreading AI benefits. However, the significant obstacles to societies benefiting from AI access raise questions about its competitiveness (relative to other development initiatives) as a means of promoting the welfare of underserved communities.

The viability of a Collaborative as a site of legitimately controlled technology also unclear: it depends on whether a balance can be struck between legitimately pursuing technologies powerful enough to positively affect international stability, and managing the proliferation of dangerous systems.

3.4 AI Safety Project: Conducting Technical Safety Research

The final model we discuss is an international collaboration to conduct technical AI safety research[26] at an ambitious scale.[27]

Tthe Safety Project would be modeled after large-scale scientific collaborations like ITER and CERN. Concretely, it would be an institution with significant compute, engineering capacity and access to models (obtained via agreements with leading AI developers), and would recruit the world's leading experts in AI, AI safety and other relevant fields to work collaboratively on how to engineer and deploy advanced AI systems such that they are reliable and less able to be misused. CERN and ITER are intergovernmental collaborations; we note that an AI Safety Project need not be, and should be organized to benefit from the AI Safety expertise in civil society and the private sector.

Functions and Motivation

The motivation behind an international Safety Project would be to accelerate AI safety research by increasing its scale, resourcing and coordination, thereby expanding the ways in which AI can be safely deployed, and mitigating risks stemming from powerful general purpose capabilities.[28] Researchers—including those who would not otherwise be working on AI safety—could be drawn by its international stature and enabled by the project's exceptional compute, engineers and model access. The Project would become a vibrant research community that benefits from tighter information flows and a collective focus on AI safety. The Project should also have exceptional leaders and governance structures that ensure its efforts are most effectively targeted at critical questions on the path to safer AI systems.

Because perceptions of AI risk vary around the world, such an effort would likely be spearheaded by frontier risk-conscious actors like the US and UK governments, AGI labs and civil society groups. In the long run, it would be important for membership to be broad to ensure its research is recognized and informs AI development and deployment around the world.[29]

[26] This could include work on understanding and evaluating characteristics of systems such as alignment/reliability and dangerous capabilities, training methods to reduce and manage these characteristics, and deployment protocols (such as system security, monitoring, accident-response) that are appropriate to different system characteristics.

[27] See, e.g., [11; 19]

[28] Being a public good, AI safety may be underfunded by default, which the Safety Project would address as a site of collective contribution.

[29] While safety-relevant insights should be publicized for international use, other innovations with commercial value can be collectively owned by or affordably licensed to member states to incentivize broad participation. See, e.g., CERN's approach to this [26].

Challenges and Risks

Pulling safety research away from frontier developers: One potential effect of this model is that it diverts safety research away from the sites of frontier AI development. It is possible that safety research is best conducted in close proximity to AI development to deepen safety researchers' understanding of the processes and systems they are trying to make safe and to ensure there is adequate safety expertise in-house. This risk could be addressed by offering safety researchers within AI labs dual appointments or advisory roles in the Project, and may become less of an issue if the *practice* of AI safety becomes institutionalized and separated from research and development.

Security concerns and model access: In order to be effective, participants in the Project need to have access to advanced AI models, which may allow them to illegally copy the model's weights, clone the model via access to its outputs [42], or understand how it could be replicated (by determining its architecture or training process). Given the importance of these assets to the business interests of frontier labs, it may be difficult to negotiate agreements where adequate model access is granted. It may also lead to the diffusion of dangerous technologies.

This issue may be addressed by restricting membership in the Safety Project and by information security measures. In particular, it may be possible to silo information, structure model access, and design internal review processes in such a way that meaningfully reduces this risk while ensuring research results are subject to adequate scientific scrutiny. Certain types of research, such as the development of model evaluations and red-teaming protocols, can proceed effectively with API access to the models, while others such as mechanistic interpretability will require access to the model weights and architectures, but may not need to work with the latest (and therefore most sensitive) models [37].

Assessment

Technical progress on how to increase the reliability of advanced AI systems and protect them from misuse will likely be a priority in AI governance. It remains to be seen whether—due to issues of model access and the allocation of experts between a Safety Project and sites of frontier AI development—an AI Safety Project will be the most effective way of pursuing this goal.

3.5 Combining Institutional Functions

We can imagine institutions taking on the role of several of the models above. For example, the Commission on Frontier AI and the AI Safety Project make an obvious pairing: a Commission could scale up research functions to supplement the synthesis and consensus-building efforts, or a Project could conduct synthesis work in the course of its activities and gradually take on a consensus-establishing role. A Frontier AI Collaborative would also likely conduct safety research, and could easily absorb additional resourcing to become a world-leading Safety Project.

4 Conclusion

This paper has outlined several reasons why the world may want to expand existing initiatives in AI governance and safety and discussed the strengths and limitations of four possible institutional models to address these needs.

To better harness advanced AI for global benefit, international efforts to help underserved societies access and use advanced AI systems will be important. A Frontier AI Collaborative that acquires and distributes AI systems could be helpful, if it can effectively enable underserved groups to take full advantage of such systems. A Commission on Frontier AI could help identify the areas where international efforts can most effectively achieve these goals, if it can prevent the politicization of its work. Relatedly, it will be important for governance approaches around the world to be coordinated,

so as to reduce frictions to innovation and access: an Advanced AI Governance Organization that sets international standards for governance of the most advanced models could facilitate this.

To manage global risks from powerful AI systems, effective AI governance regimes may be needed around the world. An Advanced AI Governance Organization that establishes governance frameworks for managing global threats from advanced systems and helps with their implementation may help internationalize effective regulatory measures, but it may be difficult to establish reliable standards if AI progress continues at the present rate, and also difficult to incentivize adoption of an Organization's standards if there is a lack of global consensus on AI risks. A Commission on Frontier AI could also support governance efforts by building scientific consensus around AI risks and their mitigation, although its task of providing a scientifically credible and internationally recognized account of a quickly changing risk landscape will be challenging, especially given the relative lack of existing scientific research on the topic. An AI Safety Project could accelerate the rate at which technical methods of mitigating AI risks are developed—provided it can overcome its efficiency and model access hurdles, and a Frontier AI Collaborative's technology might be used to increase global resilience to misused or misaligned AI systems. More speculatively, the functions of a Governance Organization and Collaborative could in some cases counteract the geopolitical factors exacerbating AI risks.

The taxonomy of functions we have presented is not exhaustive, nor do we argue that our institutional grouping is the most promising. Given the immense global opportunities and challenges presented by AI systems that may be on the horizon, the topic of international institutions for AI governance demands much greater analytical and practical attention.

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